LEO Wholesale Intelligence: Project Report

LEO Wholesale Intelligence is a data and AI tool for Leo Moda's wholesale desk. It shows the team who to call, what stock to move and what to produce, and it gives traders a 24/7 assistant that answers from live company data.

Company profile

Leo Moda is an Egyptian ready-made clothing brand, founded in Alexandria in 2014 by Eng. Abdelkhalik El Sharnouby (CEO & Founder). It is part of El-Sharnouby Group and focuses on 100% Egyptian cotton.

- **Products:** kidswear, pajamas, homewear, cotton essentials and seasonal collections (Summer and Autumn/Winter).
- **Origin:** the company started as a manufacturing partner for well-known fashion names, then built its own brand.
- **Vertical integration:** Leo Moda is involved in the whole journey: cotton cultivation in Egypt, harvesting, weaving, dyeing, cutting, sewing, packaging and delivery.
- **Promise:** comfortable everyday clothing with consistent quality, controlled at every production stage.
- **Wholesale desk:** 5 Amin Yehia St, Zizinia, Alexandria.

Business model

Leo Moda makes its own products and sells them through three channels. Wholesale is by far the largest: it brings 85.3% of net sales and 65.7% of gross profit.

Channel	Who buys	Share of net sales	Margin
Wholesale desk	1,651 traders and shop owners across Egypt	85.3%	~12%
Online store	End customers	7.7%	~35–37%
Retail store (Sidi Bishr)	End customers	7.0%	~35–37%

Total net sales in the data are 421.3M EGP with 64.9M EGP gross profit (15.4% overall margin). Traders buy in bulk at a flat ~33% discount and pay by bank transfer.

Business problem

The channel that brings the money is also the one with the least visibility and the thinnest margin. The wholesale desk works through phone calls and memory, so four problems stay hidden:

1. **Margin leak.** Every trader gets the same ~33% discount, whatever they buy. Wholesale margin is ~12% vs 35–37% in retail.
2. **Silent churn.** Good traders stop ordering and nobody notices until the revenue is gone.
3. **Stock in the wrong place.** The wholesale desk runs out of products that the stores are holding too much of.

4. **Manual inquiries.** Traders call or message to ask about price, stock and sizes, which takes the team's time and only works in office hours.

Data and approach

The analysis uses the company's sales, customers, products, inventory, locations, production, customer service and group operations files.

1. **Cleaning:** fixed date formats, seasons and missing values; checked that IDs link correctly across files; checked the stock equation (opening + in − out = closing) and split stockouts into sold vs lost sales.
2. **Trader segments (RFM):** scored every trader on recency, frequency and money spent, then grouped them into segments such as Champions, Loyal and At Risk.
3. **Overdue flag:** a trader is overdue when the gap since their last order is at least 2× their normal ordering cycle.
4. **Forecast:** a seasonal forecast per product for the next 30/60/90 days, tested on past months with WAPE before use.
5. **Stock status:** days of cover per product and location, compared with reorder points.

What I built

One end-to-end system: data notebooks feed a FastAPI backend, which serves a React dashboard and an AI assistant.

Layer	Tools	What it does
Notebooks (3)	Python, pandas, NumPy, scikit-learn, Parquet	Clean the raw files and produce 12 analytics tables (traders, segments, stock status, forecast, production plan, KPIs)
Backend API	FastAPI, Uvicorn	13 endpoints for KPIs, traders, products, stock, forecast, chat and bot orders
AI assistant	Google Gemini with function calling	Answers traders in Egyptian Arabic using 7 tools: search products, check stock per size, tiered price, clearance offers, trader profile, create order, log unmet demand
Frontend	React, Vite, Tailwind CSS, Recharts	4 pages in Leo Moda brand colors

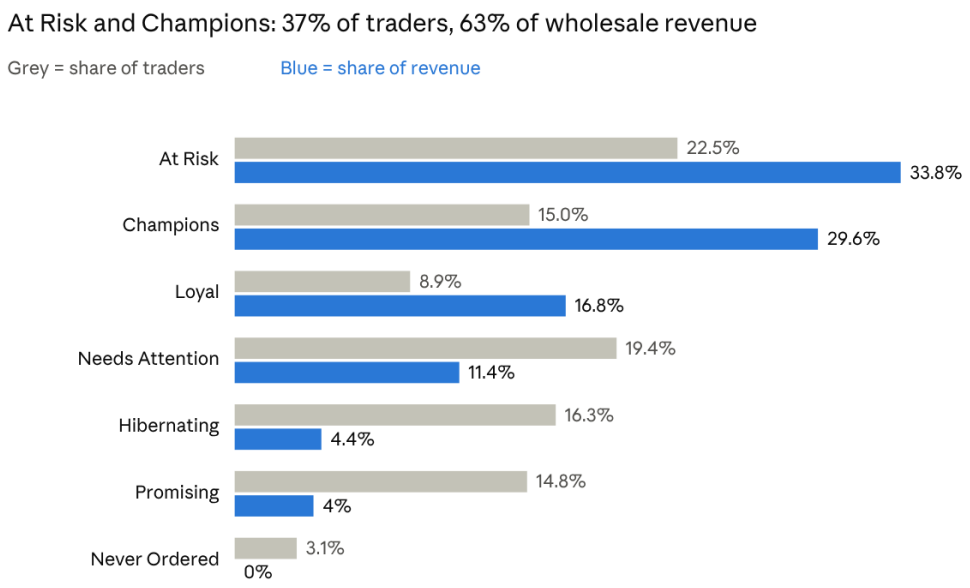
The four pages and who uses them:

- **Overview** (owner and management): headline KPIs, monthly sales by channel, key findings.

- **Traders** (wholesale sales team): segments, overdue traders, next action per trader, hidden wholesale buyers.
- **Stock & Forecast** (warehouse and production): stock health per location, transfer suggestions, 90-day forecast, production plan.
- **Wholesale Chat** (traders, via WhatsApp in production): the assistant never invents prices or stock; every number comes from a tool call.

Key insights

Insight	The number
Wholesale is the business	~5.5% of customers bring 85.3% of revenue and 65.7% of profit
The discount is the margin leak	Wholesale margin 11.9%; each 1 point of discount ≈ 5.3M EGP of profit
Revenue at risk	371 "At Risk" traders hold 33.8% of wholesale revenue
Traders slipping now	427 of 1,651 traders are overdue; 52 never ordered
Stock in the wrong place	Wholesale desk short on 49 of 59 products while stores hold 58 overstocked lines
Stock piling up	4 products hold ~52% of all stock (40,162 units), some with 500+ days of cover
Demand ahead	37,453 / 79,760 / 138,522 units in the next 30 / 60 / 90 days; peak in July, low in February
Returns	4.7% return rate; size/fit is 35.7% of returns
Production	~16-day lead time, sewing is the bottleneck, 91% fill rate



Trader segments from the analysis notebook · 1,651 traders

Where the blue bar is longer than the grey one, a small group of traders carries a large part of the revenue.

Recommendations

1. **Tie the discount to volume.** Replace the flat ~33% with tiers by order size; even 1 point back is worth ≈ 5.3M EGP.
2. **Call the At Risk and overdue traders first.** The Traders page lists them with a next action for each.
3. **Rebalance before producing.** Move about 7,300 surplus units from stores to the wholesale desk, then produce the rest.
4. **Clear the overstock through the bot.** The assistant offers clearance items once per chat, with a price floor.
5. **Give size advice before the sale** to cut size/fit returns.
6. **Connect the assistant to WhatsApp Business** so traders get answers 24/7 and the team handles fewer routine calls.

Honest limits and next steps

- **No churn prediction.** A churn model was tested and scored AUC ≈ 0.5 (no better than chance), so it was dropped. "At Risk" describes past behaviour; it does not predict.
- **CSAT is flat (~4.0) whatever the wait time.** So the bot's value is saved workload and 24/7 availability, not higher satisfaction.
- **Production volumes don't match demand in the data,** so the production file is used only for lead time, bottleneck and fill rate.
- **Per-size stock is an estimate** (stock × recent size mix).

- **Transfer rule is an assumption:** the desk targets 45 days of cover and stores keep 60.

Next steps: connect WhatsApp Business, test volume-based discount tiers with a small group of traders, and track how many At Risk traders come back after a call.